

## A Laboratory Timer using the Tang Nano 9K

Timers are essential tools of a laboratory where the duration of a process needs to be measured precisely. They typically show the time in MM:SS format, and start the down count from a set value in the range 00:01–99:59.

We would like to design such a timer, and demonstrate its operation on the Tang Nano 9K FPGA board. The timer will be set by the host computer to which the board is connected. The Tang Nano 9K board emulates a FT2232C/D/H Dual UART/FIFO IC, thus provides both a JTAG port for uploading the bit file, and a serial port through which RS232 communications can be performed.

When your design runs on the board, the display is expected to be initialized as 00:00. You will use a serial port interface program (such as PuTTY for Windows, screen for Linux and MAC, or CoolTerm for any operating system) and set the BAUD rate to 115200. Typing numbers on the serial port interface is expected to shift in the start time from right to left, i.e., if you press the key sequence 1-2-3-4, the display should show 00:01, 00:12, 01:23, and 12:34. Then pressing BUTTON0 on the board should start the countdown: 12:34, 12:33, 12:31, ..., 00:02, 00:01, 00:00, and should stop at 00:00.

You are provided with a template serial port project in *serial\_template.zip*. If everything is set properly, you should be able to see the ASCII value of the key you press on your computer on the mother board LEDs, i.e, if you press '9', LEDs should be OFF-OFF-ON-ON-ON-OFF-OFF-ON.

You may work in groups of two. Your design may be entirely in Verilog, or might be partly designed in *Digital*. Submit your entire Gowin IDE project with a name cs303\_00011111.zip if you are working alone where 00011111 is your student ID, or cs303\_00011111\_00022222.zip where 00022222 is the ID of your project partner. If you also have a part in *Digital*, also submit that with file name cs303\_00011111\_digital.zip, or cs303\_00011111\_00022222\_digital.zip.

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